



## GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

### CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING**

**COURSE CODE: CM5476**

**COURSE TITLE: ADVANCE DATABASE MANAGEMENT SYSTEM**

#### TEACHING SCHEME:

| LEVEL OF COURSE | PRERE-<br>QUISITE | WEEKLY<br>CONTACT HRS. |   |    | TOTAL<br>CREDITS | TOTAL<br>WEEKS | TOTAL CONTACT<br>HOURS |   |    |
|-----------------|-------------------|------------------------|---|----|------------------|----------------|------------------------|---|----|
|                 |                   | L                      | T | P  |                  |                | L                      | T | P  |
| V               | CM5469            | 03                     | - | 02 | 05               | 16             | 48                     | - | 32 |

#### EXAMINATION SCHEME:

| EXAMINATION SCHEME:  |      |    |     |       |                  |     |                  |
|----------------------|------|----|-----|-------|------------------|-----|------------------|
| THEORY(Marks)        |      |    |     |       | PRACTICAL(Marks) |     | TOTAL<br>(Marks) |
| ESE<br>PAPER<br>HRS. | ESE  |    | PA  | TOTAL | ESE              | PA  |                  |
| 3Hrs                 | MAX. | 70 | 30* | 100   | 25#              | 25^ | 150              |
|                      | MIN. | 28 | --- | 40    | 10               | 10  |                  |

@: Internal Assessment #: External Assessment: Practical based \$: Online Examination

(\*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

#### 1. RATIONALE:

Advanced database management system contain a comprehensive contents on various concepts related to database system database design and management discuss about parallel and distributed database systems database transaction big data management and advance in database data. The student will get a detail introduction about database administration and management the role of machine learning in big data management. This course includes study of structured and unstructured database like mongo DB SQL and XML for data management. The concept big data is used to in today's information dry driven business world of managing big data after learning the subject students will be able to ADBMS as a backend for developing database.

#### 2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

- 1) Differentiate various Database Architecture.
- 2) Use Object Oriented and Advanced XML on Database.
- 3) Manipulate data using MongoDB commands.
- 4) Use Data Warehousing Concepts.
- 5) Use Data Mining Concepts.
- 6) Use Big Data Concepts.

### 3. DETAILED CONTENTS: THEORY

| Unit                                             | Unit Outcomes (UOs)<br>(In cognitive domain)                                                                                                                                                                                                                                                                                                         | Topic and Sub-topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO No.   | Hr        | Marks     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-----------|
| <b>Unit 1<br/>Database Architecture</b>          | 1a. Describe the given client server database model.<br><br>1b. Use the given locking protocols for concurrency control<br><br>1c. Apply parallel and distributed database techniques in given situation<br><br>1d. Differentiate between parallel and distributed databases                                                                         | 1.5 Introduction to client server database model : Two Tier client server model , Three Tier client server model<br><br>1.6 Currency control technique : concurrency control protocol, lock based protocol granting of locks, two phase locking protocol<br><br>1.7 Introduction to parallel database : Parallel database system architecture ,Types of parallelism ,parallel database implementation<br><br>1.8. Introduction to distributed database :Distributed database system architecture, benefits of distributed database system | <b>1</b> | <b>08</b> | <b>12</b> |
| <b>Unit 2<br/>Object Based Databases and XML</b> | 2a. create the given object based database and XML to create the given object database using SQL<br><br>2b. Write given SQL queries using table inheritance<br><br>2c. Write given SQL queries using array and multiset<br><br>2d. Implement SQL queries to refer the given object using object identity<br><br>2e. Write XML queries on given data. | 2.1 Object based data bases overview<br><br>2.2 Complex data types<br><br>2.3 structure types and inheritance in SQL<br><br>2.4 Table inheritance<br><br>2.5 Array and multiset types in SQL<br><br>2.6 Object identity and reference types in SQL<br><br>2.7. XML introduction structure of XML data ,XML document schema, Xpath, XQuery FLOWER expression joins nested queries sorting functions ,Functions and types .                                                                                                                 | <b>2</b> | <b>10</b> | <b>14</b> |
| <b>Unit 3<br/>Advanced Database techniques</b>   | 3a. Differentiate structured and unstructured data<br>3b. Use No SQL database to solve given queries<br>3c. Use MongoDB to solve given queries.<br>3d. Differentiate SQL and NoSQL databases                                                                                                                                                         | 4.1. Structured versus Unstructured data<br>4.2. NoSQL database concepts :Types of NoSQLdatabases,NoSQL data modeling ,Benefits of NoSQL,comparison between SQL and NoSQL database                                                                                                                                                                                                                                                                                                                                                        | <b>3</b> | <b>10</b> | <b>14</b> |

|                                    |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |           |           |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-----------|
|                                    | <p>3e. Write query to execute find () functions on given data</p> <p>3f. Implement basic operations performed on MongoDB sale on given data.</p> <p>3g. Write query using aggregate() method on given data</p> | <p>system</p> <p>4.3. NoSQL using MongoDB, introduction to MongoDB shelled running the MongoDB shell MongoDB client basic operations with MongoDB shell basic data types are embedded documents.</p> <p>3.4. Quarrying with MongoDB: Find() function ,specifying which keys to return, Query criteria,ORqueries, Types specific Querying</p> <p>3.5 Aggregation introduction : Aggregation Pipeline , Aggregation using MapReduce ,Single purpose aggregation</p> |          |           |           |
| <b>Unit 4<br/>Data Warehousing</b> | <p>4a. Define Data Mart and Meta Data</p> <p>4b. Explain Architecture of Data Warehousing</p> <p>4c. Define OLAP, MOLAP, ROLAP</p>                                                                             | <p>4.1 Introduction to Data Warehouse : characteristics and types of architecture of data Warehousing</p> <p>4.2 Data Marts, Data Warehousing Life Cycle</p> <p>4.3 Data modeling (Multidimensional Database) for data Warehousing</p> <p>4.4 OLAP, MOLAP, ROLAP</p> <p>4.5 Data warehouse and views, Future open issue for data warehouse</p>                                                                                                                    | <b>4</b> | <b>06</b> | <b>08</b> |
| <b>Unit 5<br/>Data Mining</b>      | <p>5a. Analyze Given Data using Data Mining</p> <p>5b. Describe the features of</p> <p>5c.BI and BI components</p> <p>5d. Explain use of Spatial Databases in given Situation</p>                              | <p>5.1 Introduction to Data Mining : Data mining Technology and its relationship to Data warehousing, Association rules, Classification and Clustering, Application Of Data Mining</p> <p>5.2 Introduction to Business Intelligence : Features , Frameworks, Types and Approaches for Machine Learning</p> <p>5.3 Introduction to Multimedia Databases, Mobile Databases, Digital Databases</p>                                                                   | <b>5</b> | <b>08</b> | <b>10</b> |

| Unit                                      | Unit Outcomes (UOs)<br>(In cognitive domain)                                                                                                                                                              | Topic and Sub-topics                                                                                                                                                                               | CO No.   | Hr        | Marks     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-----------|
| <b>Unit 6<br/>Big Data<br/>Management</b> | 6a. Analyze the given Situation for the use of Big Data<br>6.b Describe the given architecture of Hadoop<br>6.c Explain use of Cloudera in given situation<br>6.d Explain given Features of R-Programming | 6.1 Big Data<br>6.2 Introduction to Hadoop: Building blocks, components, Hadoop Architecture, HBase, HIVE, solid –state Drive<br>6.3 Cloudera, Oracle Cloud<br>6.4 Introduction to R - Programming | <b>6</b> | <b>06</b> | <b>12</b> |

#### 4. LIST OF PRACTICALS:

| Sr No. | PRACTICAL OUTCOMES (PrOs)                                            | CO NO. |
|--------|----------------------------------------------------------------------|--------|
| 1      | Implementing Locking Protocols                                       | 1      |
| 2      | Create Database using XML attributes and Elements                    | 2      |
| 3      | Implement queries based on flower expressions and join using X query | 2      |
| 4      | Implement queries based on functions and types using X query         | 2      |
| 5      | Execute queries using structure type in SQL                          | 2      |
| 6      | Execute queries using type inheritance and table inheritance in SQL  | 2      |
| 7      | Execute queries using object identity and reference types in SQL     | 2      |
| 8      | Design and develop mongo DB queries using basic operations           | 3      |
| 9      | Implement aggregation queries using MongoDB                          | 3      |
| 10     | Import Source Data structures in Oracle                              | 4      |
| 11     | Install and configure any data mining tool (like WEKA)               | 5      |
| 12     | Install and configure RStudio tool                                   | 6      |

#### Note

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain'.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

| S. No. | Performance Indicators                            | Weightage in % |
|--------|---------------------------------------------------|----------------|
| 1.     | Installation and Configuration of Database System | 20             |
| 2.     | Coding of Queries and MongoDB Programming         | 20             |
| 3.     | Quality of Result Displayed by Queries            | 20             |
| 4.     | Answer to Sample Questions.                       | 20             |
| 5.     | Submission of report in time                      | 20             |
| Total  |                                                   | 100%           |

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Follow safety practices.
- Practice good housekeeping.
- Demonstrate working as a leader/a team member.
- Follow ethical practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1<sup>st</sup> year
- 'Organizing Level' in 2<sup>nd</sup> year and
- 'Characterizing Level' in 3<sup>rd</sup> year.

## 5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a. Prepare journal of practical.
- b. Undertake micro-projects.
- c. Prepare a chart to classify Data Structures.
- d. Prepare charts for logical representation of Data Structures.

## 6. SUGGESTED INSTRUCTIONAL STRATEGIES

These are sample strategies, which the teacher can use to accelerate the attainment of the various learning outcomes in this course:

- a. Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b. Guide student(s) in undertaking micro-projects.
- c. Demonstrate students thoroughly before they start doing the practice.
- d. Encourage students to refer different websites to have deeper understanding of the subject.
- e. Observe continuously and monitor the performance of students in Lab.
- f. 'About **10-15% of the topics/sub-topics** which is relatively simpler or descriptive in nature is to be given to the students for **self-directed learning** and assess the development of the COs through classroom presentations (see implementation guideline for details).

## 7. SUGGESTED MICRO-PROJECTS.

**Only one micro-project** is planned to be undertaken by a student assigned to him/her in the beginning of the semester. S/he ought to submit it by the end of the semester to develop the industry oriented COs. Each micro-project should encompass two or more COs which are in fact, an integration of practical's, cognitive domain and affective domain LOs. The microproject could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **16 (sixteen) student engagement hours** during the course. In the all semesters, the micro-project could be group-based ( 5-6 students) to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

- a) Develop and maintain XML database for Employee Information System.
- b) Design and Develop MongoDB Database for Library Management System.
- c) Perform Preprocessing of data using any data mining tool (WEKA).
- d) Install and Configure Hadoop.
- e) Perform database connectivity with any front end tool.
- f) Create an any XML application using any database and any programming Language (JAVA, VB.NET-ASP.NET)

- g) Implement Active Database using PL/SQL.
- h) Develop a tracker to track news from several news websites and then store them in MongoDB database.
- i) Create application for Aadhar Based Analysis using Hadoop

## 8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

The major equipment with broad specification mentioned here will usher in uniformity in conduct of experiments, as well as aid to procure equipment by authorities concerned.

| S. No. | Equipment Name with Broad Specifications                      | PrO S. No. |
|--------|---------------------------------------------------------------|------------|
| 1      | Computer System(any computer system with basic configuration) | All        |
| 2      | Any RDBMS software(MYSQL/SQL Server/Oracle/MongoDB            | 2-15       |
| 3      | Rstudio software                                              | 16         |

## 9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

| Unit No. | Unit Title                     | Marks per Unit | Distribution of Theory Marks |         |         |
|----------|--------------------------------|----------------|------------------------------|---------|---------|
|          |                                |                | R Level                      | U Level | A Level |
| 1        | Database Architecture          | 12             | 04                           | 04      | 04      |
| 2        | Object Based Databases and XML | 14             | 02                           | 04      | 08      |
| 3        | Advanced Database techniques   | 14             | 02                           | 04      | 08      |
| 4        | Data Warehousing               | 08             | 04                           | 04      | --      |
| 5        | Data Mining                    | 10             | 02                           | 04      | 04      |
| 6        | Big Data Management            | 12             | -                            | 06      | 06      |
| Total    |                                | 70             | 14                           | 26      | 30      |

*R=Remember, U=Understand, A=Apply and above (Bloom's Revised taxonomy)*

**Note:** This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

## 10. SUGGESTED LEARNING RESOURCES:

| S. No. | Title of Book                                 | Author                          | Publication                                                                  |
|--------|-----------------------------------------------|---------------------------------|------------------------------------------------------------------------------|
| 1.     | Fundamentals of Database Systems-6e           | RamezElmasri<br>ShamkantNavathe | Addison Wesley, New Delhi 2003 ISBN 13: 978-0-136-08620-9                    |
| 2.     | Database Management System Application        | Kogent Learning Solutions Inc.  | Dreamtech Press 2014,ISBN-978-93-5119-476-7                                  |
| 3.     | Database System Concepts                      | Korth Henry                     | Tata McGraw Hill Education,6 <sup>th</sup> Edition,ISBN-13:978-93-329-0138-4 |
| 4.     | Complete Reference:Mysql                      | VaswaniVikram                   | McGrawHill Education,ISBN-13:9780070586840                                   |
| 5.     | SQL,PL/SQL The Programming Language of ORACLE | Bayross Ivan                    | BPB Publications,3 <sup>rd</sup> Edition ISBN-13:978-8176569644              |

**11. SOFTWARE/LEARNING WEBSITES.**

- a. <http://nptel.ac.in/courses/106102064/1>
- b. [www.oopweb.com/algorithms](http://www.oopweb.com/algorithms)
- c. [www.studytonight.com/data-structures/](http://www.studytonight.com/data-structures/)
- d. [www.cs.utexas.edu/users](http://www.cs.utexas.edu/users)
- e. [www.liscs.wssu.edu](http://www.liscs.wssu.edu)
- f. <http://www.academictutorials.com/data-structures>
- g. <http://www.sitebay.com/data-structure/c-data-structure>
- h. <http://www.geeksforgeeks.org/data-structures/>

**12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:**

| SR. NO. | NAME          | DESIGNATION                         | INDUSTRY/INSTITUTE         |
|---------|---------------|-------------------------------------|----------------------------|
| 1       | S. S. CHAVHAN | Lecturer in Computer Engineering.   | Govt. Polytechnic Amravati |
| 2       | A.P.JANE      | Lecturer in Information Technology. | Govt. Polytechnic Amravati |
| 3       | Dr. P.P.Karde | Lecturer in Information Technology. | Govt. Polytechnic Amravati |

Government Polytechnic Amravati, Programme Board of Studies Computer Engineering has approved the above course curriculum on 30/12/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN  
PROGRAMME BOARD OF STUDIES,  
COMPUTER ENGINEERING  
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 06/02/2021

The Governing Body has approved the above course curriculum on 13/08/2021

## 11. CDC Formats 1-13



### GOVERNMENT POLYTECHNIC, AMRAVATI.

(An Autonomous Institute of Govt. of Maharashtra)

#### CURRICULUM DEVELOPMENT CELL

PROFORMA –CDC-1

w.e.f. 2018-19

#### TEACHING PLAN (TP)

Academic Year/Term: .....

Program: ..... Shift: I/II

Course: ..... Course Code: .....Name of Faculty: .....

| Unit<br>no/Allo<br>cated<br>Hrs | Details of topic/subtopic | CO<br>no. | UO<br>no. | Target<br>Hrs | Plan dates /<br>weeks<br>From--To | Actual dates<br>/weeks<br>From-- To | Remarks |
|---------------------------------|---------------------------|-----------|-----------|---------------|-----------------------------------|-------------------------------------|---------|
|                                 |                           |           |           |               |                                   |                                     |         |

Signature of the Faculty

Signature of the HOD